NMC 2010 A-LEVEL PAPER 1 QUESTIONS

SECTION A

1. Evaluate the sum to infinity

$$\frac{1}{7}+\frac{2}{7^2}+\frac{1}{7^3}+\frac{2}{7^4}+\ldots.$$

2. Two candles of the same height are lighted at the same time. The first is consumed in 4 hours and the second in 3 hours. Assuming that each candle burns at a constant rate. In how many hours after being lighted was the first candle twice the height of the second?

3. The sum of seven integers is -1. What is the maximum number of the seven integers that can be larger than 13?

4. The marked price of a coat was 40% less than the suggested retail price. Hasna purchased the coat for half the marked price at a promotion on the fiftieth anniversary sale. What percent less than the suggested retail price did Hasna pay?

5. In a group of cows and chicken, the number of legs was 14 more than twice the number of heads, find the number of cows in the group.

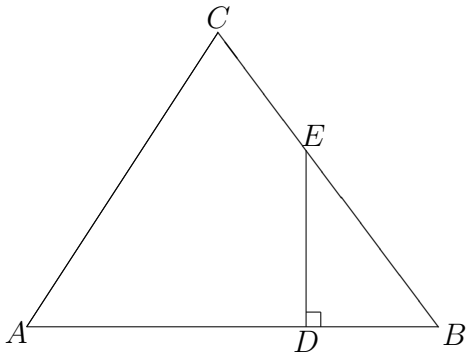
6. In solving a problem that reduces to a quadratic equation one student makes a mistake only in the constant term of the equation and obtains 8 and 2 for the roots. Another student makes a mistake only in the coefficient of the first degree term and finds -9 and -1 for the roots. write down the correct equation.

7. A total of 28 handshakes were exchanged at the conclusion of a party. Assuming that each participant was equally polite towards all the others. How many people attended the party?

8. If a worker receives a 20% decrease in wages. By what percentage should his current pay be increased in order to regain his original wage before the decrease?

9. Find the value of x if,
$$\sqrt{5x-1}+\sqrt{x-1}=2?$$

10. In the figure below, it's given that angle $C=90^\circ$, $\overline{AD}=\overline{DB}$, $\overline{DE}\perp\overline{AB}$. $\overline{AB}=20cm$ and $\overline{AC}=12cm$. Find the area of the quadrilateral $ADEC$.



SECTION B : 10 **marks each**

11. (a) The sum of three numbers is 98, the ratio of the first to the second is $\frac{2}{3}$, and the ratio of the second to the third is $\frac{5}{8}$. Find the second number?

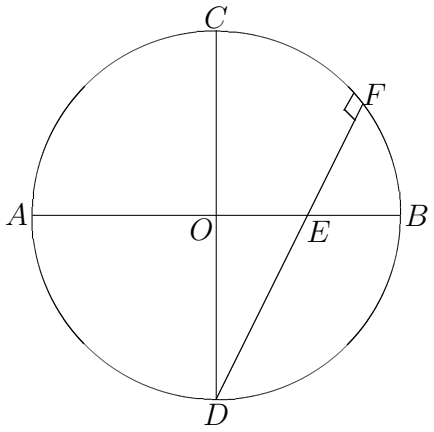
    (b) A farmer divides his herd of n cows among his four sons so that one son gets one half of the herd, a second son gets one fourth, a third gets one fifth and the fourth son gets 7 cows. Find the total number of cows and how many each son got?

12. A function f from integers to integers is defined as follows, $f(n) = n + 3$ if n is odd and $f(n) = \frac{n}{2}$ if n is even. Suppose k is odd and $f(f(f(k))) = 27$, what is the sum of the digits of k?

13. A wheel with a rubber tire has an outside diameter of 25 inches. Assuming that it's radius is reduced by a quarter of an inch. Find the percentage increase in the number of revolutions of the wheel in one mile.

14. Tom, Dick, and Harry started out on a 100 mile journey. Tom and Harry went by automobile at the rate of $25mph$ while Dick walked at the rate of $5mph$. After a certain distance, Harry got off and walked on at $5mph$ while Tom went back for Dick and got him to the destination at the same time that Harry arrived. Find the number of hours spend on the journey?

15. In the figure below, $\overline{AB}$ and $\overline{CD}$ are diameters of the circle with center O, $\overline{AB} \perp \overline{CD}$, and chord $\overline{DF}$ intersects $\overline{AB}$ at E. If $DE = 6cm$ and $EF = 2cm$ then find the area of the circle.



NMC 2010 A-LEVEL PAPER 1 SOLUTIONS

SECTION A

1.

$$\begin{aligned}\frac{1}{7}+\frac{2}{7^2}+\frac{1}{7^3}+\frac{2}{7^4}+\dots &= \left(\frac{1}{7}+\frac{1}{7^3}+\frac{1}{7^5}+\dots\right)+\left(\frac{2}{7^2}+\frac{2}{7^4}+\frac{2}{7^6}+\dots\right)\\ &= \frac{\frac{1}{7}}{1-\left(\frac{1}{7}\right)^2}+\frac{2\frac{2}{7^2}}{1-\left(\frac{1}{7}\right)^2}\\ &= \frac{7}{48}+\frac{2}{48}\\ &= \frac{9}{48}=\frac{3}{16}.\end{aligned}$$

2. Let x be the height of the two candles. The burning rate of A is $\frac{x}{4}$ and the burning rate of B is $\frac{x}{3}$. At a time t when the height of A is twice the height of B, A will have lost $\frac{tx}{4}$ of its height and B will have lost $\frac{tx}{3}$ of its height. The remaining heights of candles A and B are $x-\frac{tx}{4}$ and $x-\frac{tx}{3}$ respectively. The remaining heights of the two candles satisfy the equation $x-\frac{x}{4}t=2\left(x-\frac{x}{3}t\right)$ or $1-\frac{1}{4}t=2\left(1-\frac{1}{3}t\right)$ from which t is found to be $2\frac{2}{5}$ hours.

3. It is not possible for all the seven integers to be greater than 13. If six of them where each 14, then the seventh could be $-(6\times 14)-1$ so that the sum is -1. Thus, the maximum number larger that 13 is 14.

4. Suppose that the suggested retail price was P. Then, the marked price was $0.6P$ and Hasna bought it at half the marked price which is $0.3P$. This means that Hasna bought it at 70% less than the suggested retail price.

5. Let the number of cows be a and the number of chicken be b. The number of legs are $4a+2b$ and the number of heads is $a+b$. So,we have $4a+2b=2(a+b)+14$ or $2a=14$ or $a=7$ implying that there were 7 cows in the group.

6. Let the quadratic equation be of the form $ax^2-bx+c=0$. Sum of the roots is $\frac{b}{a}$ and product of the roots is $\frac{c}{a}$. The first student makes a mistake only in the constant term implying that the sum of the root $\frac{b}{a}=8+2=10$ is correct. This implies that $b=10a$. The second students makes a mistake only in the linear coefficient implying that the product $\frac{c}{a}=-9\times -1=9$ is the correct one. This implies that $c=9a$. The equation is therefore $ax^2-10ax+9a=0$ *or* $x^2-10x+9=0$.

7. Let n be the total number of people at the party. If each person exchanged a handshake with a different person once, then we have $\begin{pmatrix} n \\ 2 \end{pmatrix} = \frac{n(n-1)}{2}$ handshakes in all. Given that there were 28 handshakes, in all, we solve $\frac{n(n-1)}{2} = 28$ or $n^2 - n - 56 = 0$ implying that $n = 8$ or -7. We ignore -7 and take $n = 8$.

8. Let the original wage be w. After the decrease, his new wage is $w_1 = 80\%w$. If this wage bill is increased by $I\%$, the new wage bill will be $w_2 = 80\% \times (100 + I)\%$. Since the new bill is equal to the first bill, we solve for $80\% \times (100 + I)w\% = w$ implying that $I = 25$. Therefore, we have to increase the current wage by 25% to get the original wage.

9. The equation $\sqrt{5x-1} + \sqrt{x-1} = 2$ can be written as $\sqrt{5x-1} = 2 - \sqrt{x-1}$. Squaring gives

$$\begin{aligned} 5x - 1 &= 4 - 4\sqrt{x-1} + x - 1 \\ \Rightarrow\ 4x - 4 &= -4\sqrt{x-1} \\ \Rightarrow\ x - 1 &= -\sqrt{x-1} \\ \Rightarrow\ (x-1)^2 &= x - 1 \\ \Rightarrow\ (x-1)^2 - x - 1 &= 0 \\ \Rightarrow\ x^2 - 3x + 2 &= 0 \\ \Rightarrow\ x = 1 &\text{ or } x = 2. \end{aligned}$$

   Of the two solutions, only $x = 1$ satisfies the equation $5x - 1 = 4 - 4\sqrt{x-1} + x - 1$. So $x = 1$ is the solution.

10. Since $\overline{AB} = 20cm$ and $\overline{AC} = 12cm$, then, $\overline{BC} = 16cm$. The height CA of $\triangle ACB$ is $16 \tan B = 12$ which implies that $\tan B = \frac{12}{16}$. Likewise, the height DE of $\triangle EDB$ is $\overline{DE} = \overline{DB} \tan B = 10 \times \frac{12}{16}$. Therefore,

$$\begin{aligned} \text{Area of quadrilateral } ADEC &= \text{Area of } \triangle ACB - \text{Area of } \triangle EDB \\ &= \frac{1}{2}\overline{CB} \times \overline{AC} - \frac{1}{2}\overline{DB} \times \overline{DE} \\ &= \frac{1}{2} \times 16 \times 12 - \frac{1}{2} \times 10 \times 10 \times \frac{12}{16} \\ &= 58\frac{1}{2}. \end{aligned}$$

11. (a) Let the numbers be x, y and z. We are given that $x + y + z = 98$, $\frac{x}{y} = \frac{2}{3}$ and $\frac{y}{z} = \frac{5}{8}$. The second equation implies that $y = \frac{3}{2}x$ and the third equation implies that $z = \frac{8}{5}y = \frac{8}{5} \times \frac{3}{2} = \frac{12}{5}x$.

So, $x+y+z=98$ implies that $x+\frac{3}{2}x+\frac{12}{5}x=98$ which in turn imply that $\frac{49}{10}x=98$ from which we get $x=20$. The second number is therefore equal to $\frac{3}{2}x=30$.

(b) From the information given in the question, we have, $n=\frac{1}{2}n+\frac{1}{4}n+\frac{1}{5}n+7$ implying that $n=140$. Therefore, the first son gets $\frac{n}{2}=70$ cows, the second gets $\frac{n}{4}=35$ cows, the third gets $\frac{n}{5}=28$ cows and the fourth gets 7 cows.

12. Since k is odd, $f(k)=k+3$. Since $k+3$ is even,

$$f(f(k))=f(k+3)=\left(\frac{k+3}{2}\right).$$

If $\left(\frac{k+3}{2}\right)$ is odd, then,

$$f(f(f(k)))=f\left(\frac{k+3}{2}\right)=\frac{k+3}{2}+3=\frac{k+9}{2}=27$$

which, on solving gives $k=45$. This is not possible because

$$f(f(f(45)))=f(f(48))=f(24)=12.$$

Hence, $\left(\frac{k+3}{2}\right)$ is even and

$$f(f(f(k)))=f\left(\frac{k+3}{2}\right)=\frac{k+3}{4}=27$$

implying that $k=105$. Checking, we find that

$$f(f(f(105)))=f(f(108))=f(54)=27.$$

The sum of the digits of k is $1+0+5=6$.

13. Let D be the diameter of the wheel before it is reduced. The number of revolutions N_1 before the diameter is reduced are $N_1=\frac{1\,\text{mile}}{\pi D}$. The number of revolutions N_2 after the diameter is reduced by $\frac{1}{2}$ an inch is $N_2=\frac{1\,\text{mile}}{\pi\left(D-\frac{1}{2}\right)}$. The number of revolutions increase by

$$N_2-N_1 = \frac{1}{\pi\left(D-\frac{1}{2}\right)}-\frac{1}{\pi D}=\frac{1}{2\pi D\left(D-\frac{1}{2}\right)}.$$

$$\frac{N_2-N_1}{N_1} = \frac{1}{2\left(D-\frac{1}{2}\right)}=\frac{1}{2\times 24\frac{1}{2}}=\frac{1}{49}.$$

The revolutions increase by $\frac{100}{49} = 2\%$.

14. Let t_1, t_2 and t_3 be the number of hours respectively that the car travels forwards, back to pick up, then forward to the destination. During this time, Harry, traveled forwards a distance given by $25t_1 + 5t_2 + 5t_3 = 100$ while Dick traveled forwards a distance of $5t_1 + 5t_2 + 25t_3 = 100$. On the other hand, Tom traveled a distance of $25t_1 + 25t_3$ forwards with an overlap for a distance of $25t_2$. Therefore, net forward distance traveled by tom is $25t_1 + 25t_3 - 25t_3 = 100$. We then solve the following equations

$$\begin{aligned} 25t_1 + 5t_2 + 5t_3 = 100 &\Rightarrow 5t_1 + t_2 + t_3 = 20 \quad (15)\\ 5t_1 + 5t_2 + 25t_3 = 100 &\Rightarrow t_1 + t_2 + 5t_3 = 20 \quad (16)\\ 25t_1 + 25t_3 - 25t_3 = 100 &\Rightarrow t_1 - t_2 + t_3 = 4 \quad (17) \end{aligned}$$

Equation (16)+ (17) and Equation (15)- (16)yield

$$\begin{aligned} t_1 + 3t_3 &= 12\\ t_1 - t_3 &= 0 \end{aligned}$$

whose solution is $t_1 = t_3 = 3$. Using Equation (15) generates $t_2 = 2$. Hence, $t_1 + t_2 + t_3 = 8$ hours.

15. Draw segment FC. Note that $\angle CFD$ is a right angle since arc CFD is a semi circle. Then, the right angled triangles DOE and DFC are similar. If we let $DO = r$. Then,

$$\frac{\overline{DO}}{\overline{DF}} = \frac{\overline{DE}}{\overline{DC}} \Rightarrow \frac{r}{8} = \frac{6}{2r} \Rightarrow 2r^2 = 48 \Rightarrow r^2 = 24.$$

Then, the area of the circle is $\pi r^2 = 24\pi cm^2$.